

28. When ethyl ethanoate, $\text{CH}_3\text{C}^{16}\text{O}_2\text{C}_2\text{H}_5$, undergoes acid hydrolysis in a solution containing H_2^{18}O ,

atoms of the ^{18}O isotope appear in the product as $\text{CH}_3\text{C}(=\text{O})^{18}\text{OH}$ and **not** as $\text{C}_2\text{H}_5^{18}\text{OH}$.

What is a correct deduction from this observation?

A A $^{16}\text{O}-\text{C}$ bond is stronger than a $^{18}\text{O}-\text{C}$ bond.

B The $\text{O}-\text{C}$ bond in the $\text{O}-\text{C}_2\text{H}_5$ group of the ester is broken.

C The $\text{C}-\text{O}$ single bond in the $-\text{C}(=\text{O})\text{O}-$ group of the ester is broken.

D The H_2^{18}O attracts a $\delta+$ carbon atom in the C_2H_5 group.